ARJUN KRISHNAN SURESH

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SUMMARY

AI & Data Science undergraduate (B.Tech, expected 2028) building production LLM systems: multi-agent voice pipelines, hybrid RAG retrieval, and offline evaluation harnesses, each measured against latency budgets, pytest suites, and accuracy baselines rather than notebook demos. Co-author on a deep-learning research paper in battery state-of-charge prediction. Seeking an AI/ML engineering internship.

EDUCATION

Amrita Vishwa Vidyapeetham, School of Artificial Intelligence, Faridabad, India **Aug 2024 - May 2028**

B.Tech in Artificial Intelligence & Data Science | CGPA: 8.5/10

Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Database Management Systems, Computer Networks, Cryptography & Network Security, IoT, Robotics

EXPERIENCE

Secretary, Amrita Council of Sports and Excellence **May 2026 - Present**

- Directed logistics and operations for 5 university-wide sports and student-excellence events serving 200+ participants, owning scheduling, venue booking, and on-day execution.
- Secured INR 10,000 in sponsorship and launched "Respawn Rush," a campus-wide BGMI esports tournament with 11 teams (44 players), establishing esports as a recognized competitive event within the council.

TECHNICAL PROJECTS

CommerceMind VoiceCare AI (Flagship): Voice-First Multilingual Customer Support **2026**

Next.js, FastAPI, LangGraph, Gemini 2.5, Bhashini, ChromaDB, PostgreSQL (Neon), Redis (Upstash)

- Delivered spoken issue resolution in under 10 seconds for 9 Indian languages by architecting a 9-agent LangGraph pipeline (speech-to-text, intent detection, policy RAG, resolution generation, text-to-speech) streamed in real time over WebSocket.
- Verified reliability with 57 passing unit tests, P50/P95/P99 latency monitoring, and 90% average intent-detection accuracy measured on a 35-conversation evaluation set, surfaced through an admin dashboard for escalation and resolution analytics.

Production-Ready RAG System: Hybrid Retrieval Engine **2026**

FastAPI, ChromaDB, sentence-transformers, Streamlit, pytest

- Raised retrieval Precision@5 from a 68% cosine-similarity baseline to 84% by fusing BM25 keyword search with dense retrieval via Reciprocal Rank Fusion and MS MARCO cross-encoder reranking, keeping the LLM grounded in consistently relevant context.
- Cut per-token inference cost to zero with a provider-agnostic switch between the OpenAI SDK and a local Ollama instance (Llama 3, Mistral), running embeddings and reranking fully offline with deduplication and SSE streaming covered by a 48-test pytest suite.

Deep Learning Battery Management System for IoT Nodes **2025**

Research Paper (Co-author) - Python, TensorFlow, Keras, BiLSTM, SMOTE, scikit-learn

- Co-authored a BiLSTM battery-management framework using dual-phase temporal modeling of charge and discharge cycles, reaching 0.9953 R^2 on held-out state-of-charge prediction and 94.18% energy savings.
- Benchmarked ML-based battery health estimation, remaining useful life (RUL) prediction, and operating-mode classification against conventional threshold-based control on 19,735 IoT sensor readings, split chronologically 80/20 to prevent look-ahead leakage.

ADDITIONAL PROJECTS

- **Natural Language to App Schema Compiler (2026):** Compiled plain-English specifications into validated app schemas through a 5-stage Gemini pipeline, hitting a 90% pass rate on a 20-test automated evaluation harness with a repair engine that retries only the failed stage.
- **Local LLM Benchmarking Assistant (2026):** Scored 6 open-weight models on 40 tasks fully offline under 3 independent graders (exact match, LLM-as-judge, and sandboxed code execution), recording per-model latency and throughput.
- **Task Manager Agent (2026):** Coordinated scheduling, note-taking, and day planning across specialized Google ADK sub-agents, deployed as a managed service on Cloud Run.

LEADERSHIP & ACHIEVEMENTS

- **Competitive Athletics:** Placed 3rd for Amrita Vishwa Vidyapeetham at an inter-collegiate badminton tournament in Chennai, and competed in multiple sports for the B.Tech AI & Data Science cohort.
- **Google Build with AI Workshop Series (2026):** Built agentic AI applications over 3 hands-on workshops covering Google ADK, MCP servers for BigQuery and Google Maps, and Gemini-powered applications, run by Orca AI Club with Google.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 AI Foundations Associate, Oracle University (Oct 2025 - Oct 2027) | Google Cloud Gen AI Academy APAC 2026, Cohort 1, Google Cloud and Hack2skill (Apr 2026) | Python Foundation Certification & Artificial Intelligence Primer, Infosys Springboard (Mar 2026)

SKILLS

Languages & Tools: Python, TypeScript/JavaScript, C++, SQL, MATLAB, Git, Docker
AI / ML: TensorFlow, Keras, scikit-learn, NumPy, Pandas, BiLSTM, SMOTE, Conversational AI & NLP
LLMs & Agents: LangGraph, RAG (hybrid retrieval, RRF, cross-encoder reranking), Gemini 2.5, GPT, Ollama (Llama 3, Mistral), Agentic AI (ADK, MCP), LLM-as-judge evaluation, vector databases, prompt engineering
Backend & Data: FastAPI, Flask, PostgreSQL (Neon), MySQL, Redis (Upstash), SQLAlchemy, ChromaDB, REST APIs, WebSockets, pytest
Frontend & Infra: Next.js, React, Tailwind CSS, Framer Motion, Streamlit, Google Cloud, OCI, Vercel
Spoken Languages: English (Fluent), Hindi (Proficient), Malayalam (Native)